Computational Bioinformatics

Reference-based RNA-seq and plant GPX1 phylogeny

Undergraduate research artefacts | Public edition | 18 October 2026

I completed two coursework exercises to learn how biological sequence data become interpretable results. The first used Galaxy for a Drosophila RNA-seq analysis; the second used MEGA12 for protein phylogeny. This report preserves selected outputs and the reasoning behind the analysis, including problems encountered during the work.

Study 1: Pasilla depletion in Drosophila

The expression analysis compares four untreated samples with three samples treated by Pasilla RNAi. I included sequencing layout as a technical covariate. The source study is GEO GSE18508, with tutorial inputs from Zenodo record 6457007.

Sample	Treatment	Sequencing
GSM461176	untreat	single
GSM461177	untreat	paired
GSM461178	untreat	paired
GSM461182	untreat	single
GSM461179	treat	single
GSM461180	treat	paired
GSM461181	treat	paired

Study 2: Plant GPX1 protein trees

I compared Neighbor-Joining and Maximum Likelihood tree views from MEGA12. The retained ML caption records 10 amino-acid sequences, 245 sites, the JTT model, and 500 bootstrap replicates. The figures contain missing tip labels, so the public interpretation is deliberately limited to the evidence that remains visible.

These are two separate educational analyses. This archive contains methods, selected data, original tree images, and an attributed workflow. It does not contain all files required to rerun either study end to end.

1. RNA-seq workflow and troubleshooting

The genome-based workflow

I organized subsampled paired-end FASTQs for GSM461177 and GSM461180 and completed Cutadapt trimming. The intended next steps were STAR mapping to Drosophila Release 6 and featureCounts counting with Drosophila_melanogaster.BDGP6.32.109_UCSC.gtf.gz. Falco and STAR jobs failed. For differential expression, I continued with the seven featureCounts tables supplied by the tutorial.

Step	Input and purpose	Recorded outcome
Falco / MultiQC	Read quality assessment and summary	Falco failed; no usable QC summary retained
Cutadapt	Paired reads; trim and filter	Completed
STAR	Trimmed reads, reference genome, annotation	Failed; no mapping rate reported
featureCounts	BAM plus annotation; count genes	No counts produced from this failed mapping branch
DESeq2	Seven tutorial count files and sample factors	Completed after tag repair
goseq / KEGG	Selected genes, lengths, category mappings	GO and KEGG outputs recorded
heatmap2 / Pathview	Normalized counts or expression changes	Outputs recorded; originals not recovered here

A metadata error before DESeq2

The collection identifiers had four components, for example GSM461176_untreat_single_featureCounts. A three-group expression, (.*)_(.*)_(.*), absorbed the treatment token into the first group. The resulting tags contained single, paired, and featureCounts, but not treat/untreat. Changing the parser to (.*)_(.*)_(.*)_(.*) restored both treatment and sequencing tags.

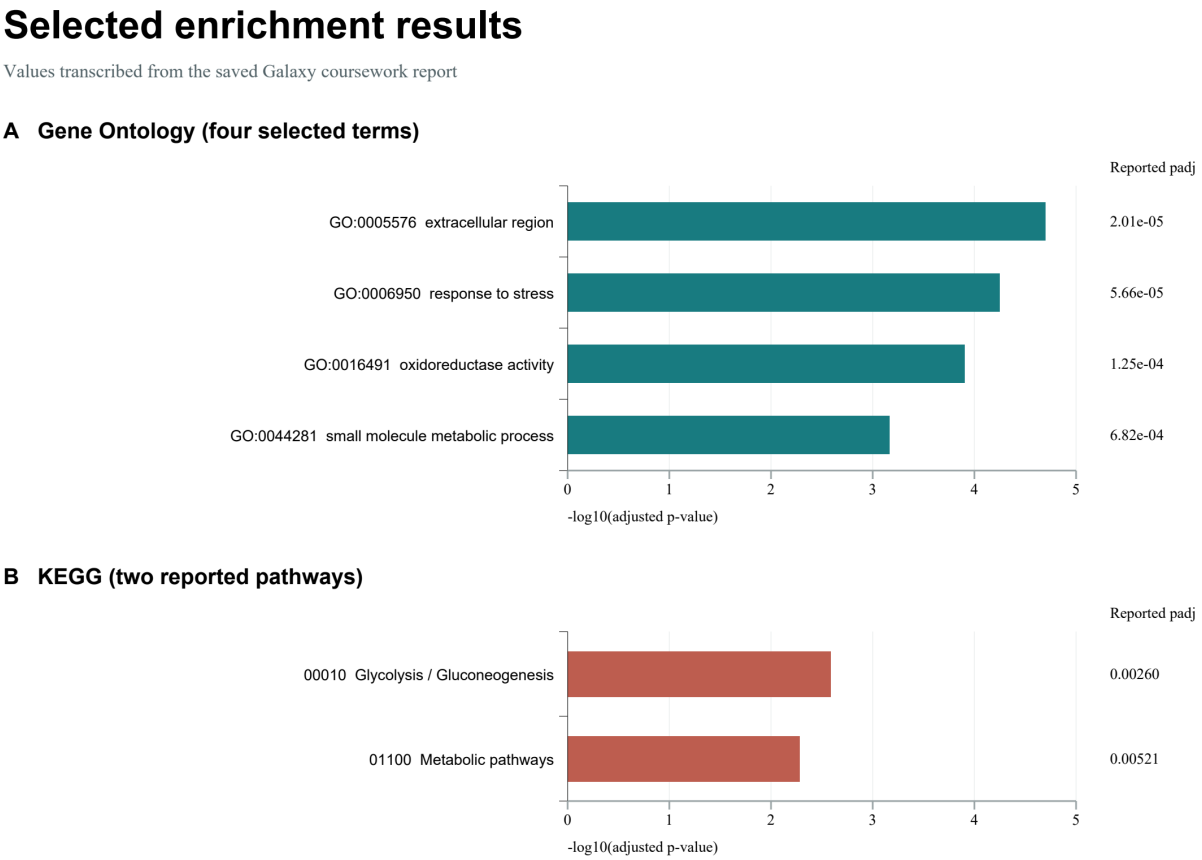
Statistical design

The model can be expressed as ~ Sequencing + Treatment, comparing treat against untreat. DESeq2 uses raw integer counts, size-factor normalization, negative-binomial models, and multiple-testing adjustment. The normalized-count matrix is an output for interpretation, not a replacement input for the count model.

Galaxy displayed 23,932 rows in the differential-expression table and 23,933 in normalized counts. These are dataset row counts, not numbers of significant genes. Exact header handling has not been recounted in this archive.

2. Selected GO and KEGG results

The saved report and dataset preview record 60 filtered GO rows; the complete GO table was displayed as 12,215 rows. The figure below uses four selected GO terms and the two reported KEGG rows. Values are rounded transcriptions from the coursework, rather than a new enrichment calculation.



Identifier	Category / reported counts	Adjusted p-value
GO:0005576	CC	2.01e-05
GO:0006950	BP	5.66e-05
GO:0016491	MF	1.25e-04
GO:0044281	BP	6.82e-04
00010	14/47	0.00260
01100	96/871	0.00521

goseq accounts for selection bias associated with gene length. The tutorial uses a 0.05 adjusted-p-value filter; the complete submitted parameter export and background list are not included here. Enrichment describes association with a category and does not establish pathway activation or causality.

The reported Pathview outputs were dme00010 and dme03040. Visualization of dme03040 does not make it one of the two significantly enriched KEGG rows (00010 and 01100).

3. Archived Neighbor-Joining trees

The coursework describes MEGA12 amino-acid NJ analysis, Poisson correction, and 500 bootstrap replicates. The following images are the original embedded figures. Unlabeled tips are preserved, so a full accession-to-tip mapping cannot be reconstructed from them.

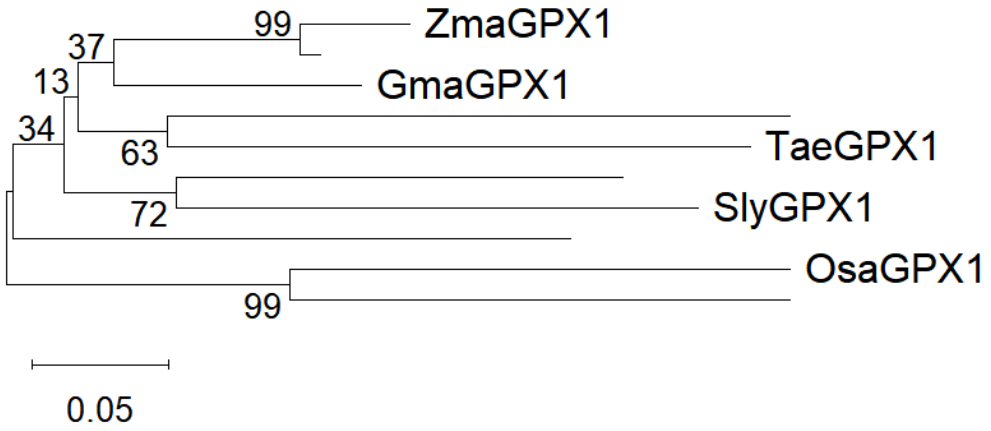


Figure 1. NJ, first displayed species set. Labels include maize, soybean, wheat, tomato, and rice. The 99 near ZmaGPX1 is next to an unlabeled tip pair; it does not support the earlier prose claim of a maize-soybean pair at 99. Scale: 0.05 substitutions per site.

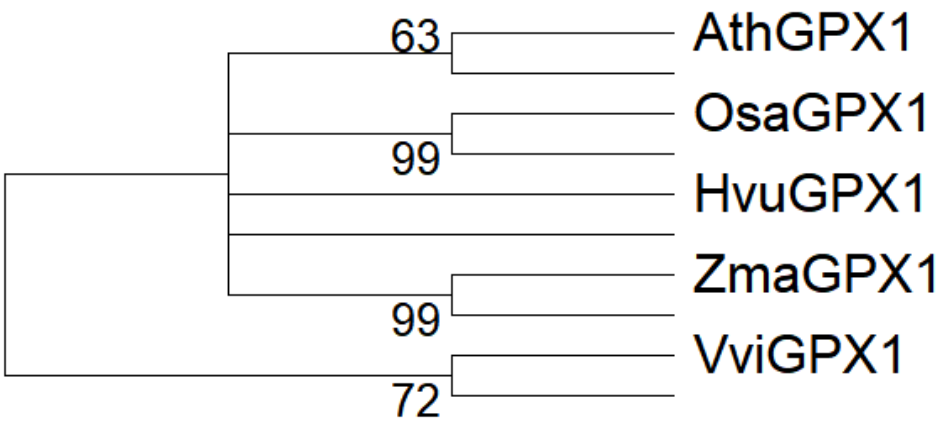


Figure 2. NJ, second displayed species set. Labels include Arabidopsis, rice, barley, maize, and grape. The image has missing tip names and no distance scale. I do not infer the missing identities.

4. Maximum Likelihood and bootstrap

The original MEGA-generated caption records JTT, 500 bootstrap replicates, 10 sequences, 245 sites, and a highest displayed log likelihood of -2498.63. This direct software record corrects the coursework prose that referred to five sequences. Five visible species labels do not imply five input sequences.

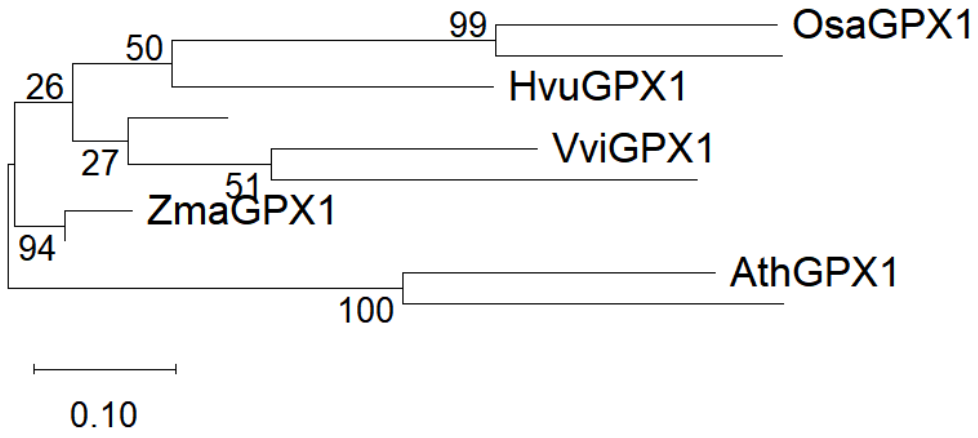


Figure 3. ML original tree. Values 26 and 27 indicate weakly supported splits. The 99 beside OsaGPX1 belongs to its pair with an unnamed tip; the broader branch containing HvuGPX1 is marked 50. The image does not support a claim of rice-barley support at 99. Scale: 0.10 substitutions per site.

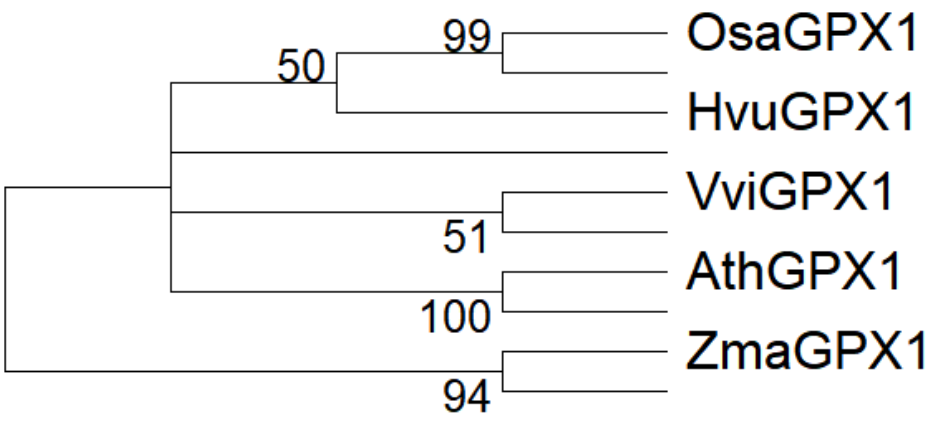


Figure 4. ML bootstrap consensus. Bootstrap values concern resampled alignment splits, not the probability that a species relationship is true. Horizontal lengths in this unscaled view should not be interpreted as evolutionary distances.

5. Interpretation, limitations, and sources

What I learned

The DESeq2 failure showed me why sample names and metadata deserve the same attention as statistical tools. Tracing the problem to the tagging step was more useful than changing DESeq2 versions. The phylogeny exercise taught me to read support values at the exact branch and to check that every tip has an identity before drawing a biological conclusion.

What the archive can support

It documents a seven-sample expression design, successful downstream outputs recorded in Galaxy, a metadata repair, selected functional terms, and actual MEGA tree images. It also demonstrates the need to separate observed results from stronger biological claims. These are practical foundations for further work in computational biology and AI for Science.

Limits and a reproducible next iteration

STAR and Falco failed, and the counts used downstream came from GTN. The later Galaxy IQ-TREE 2.4.0 attempt, with 1,000 bootstrap replicates reported as an intended setting, also failed. Its cause cannot be established from the retained screenshot. I have not claimed independent validation by IQ-TREE.

The original Galaxy session was no longer authenticated during archiving. Full result matrices, standalone MA plots and heatmaps, a repaired downstream workflow export, aligned proteins, and Newick trees are not included. A next iteration would recover these files, restore unique sequence labels, document all accessions and model settings, and rerun the incomplete branches.

Sources and attribution

Galaxy Training Network. Reference-based RNA-Seq data analysis. The included upstream workflow retains its creators and MIT license.

Brooks et al. (2011). Conservation of an RNA regulatory map between Drosophila and mammals. Data: GEO GSE18508; training files: Zenodo 6457007.

Love, Huber and Anders (2014). DESeq2. doi:10.1186/s13059-014-0550-8.

Young et al. (2010). Gene ontology analysis for RNA-seq: accounting for selection bias. doi:10.1186/gb-2010-11-2-r14.

MEGA. MEGA12 Maximum Likelihood documentation. Software-caption values were checked against the retained coursework images.

Repository: github.com/Rianzhan/computational-bioinformatics-rnaseq-phylogeny

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